

BUSINESS VALUE THROUGH INTEGRATIVE ANALYTICS

Telco Customer *Churn* Retention

Optimizing intervention budget across 7,000+
at-risk customers

TEAM

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EXECUTIVE SUMMARY

Prioritize high-CLTV customers with a tiered outreach program

~90%

Model AUC

Random Forest — leakage-free, satisfaction score excluded after diagnosis

+26%

IP lift vs. random targeting

Integer program outperforms all heuristics on expected retention value

\$7.5K

Budget deployed optimally

3-tier program: Email (\$5), Phone (\$15), Premium Package (\$50)

RECOMMENDATION

Deploy IP-driven 3-tier retention program. Focus Premium Package on Cluster 0 (highest churn, high CLTV). Risk-adjusted model remains robust across $\lambda = 0-2$.

BUSINESS PROBLEM + DATA

Churn is expensive — and predictable

THE PROBLEM

~27%

Annual churn rate in U.S. telecom

Acquiring a new customer costs 5–7× more than retaining one. The question is not *whether* to intervene — but *who* to intervene with, and *how*.

KEY ASSUMPTION

Intervention effectiveness (η) is fixed per tier. CLTV is treated as a reliable proxy for customer lifetime value.

DATASET

7,044

Customers

33

Features

4

Clusters

- Telco Customer Churn (IBM)

Source

- satisfaction_score excluded

Corr -0.75 → leakage

- Contract, tenure, monthly charge

Top predictors

- K-Means: 4 customer segments

6 behavioral features

OPTIMIZATION MODEL

Assign each customer to the right intervention tier

DECISION

For each customer i , assign exactly one intervention tier j — or no intervention at all.

OBJECTIVE

Maximize total **expected retention value** = probability of churn $\times$ retention rate $\times$ customer CLTV – tier cost

CONSTRAINTS

- 1. Total spend $\leq$ \$7,500 budget
- 2. At most one tier per customer
- 3. Safety: $p_{\text{churn}} > 0.80 \rightarrow$ must be contacted

TIER	COST	RETENTION H
Email / SMS	\$5	10%
Phone Outreach	\$15	25%
Premium Package	\$50	45%

Value per customer-tier assignment

$$v_{ij} = \hat{p}_i \times \eta_j \times CLTV_i - c_j$$

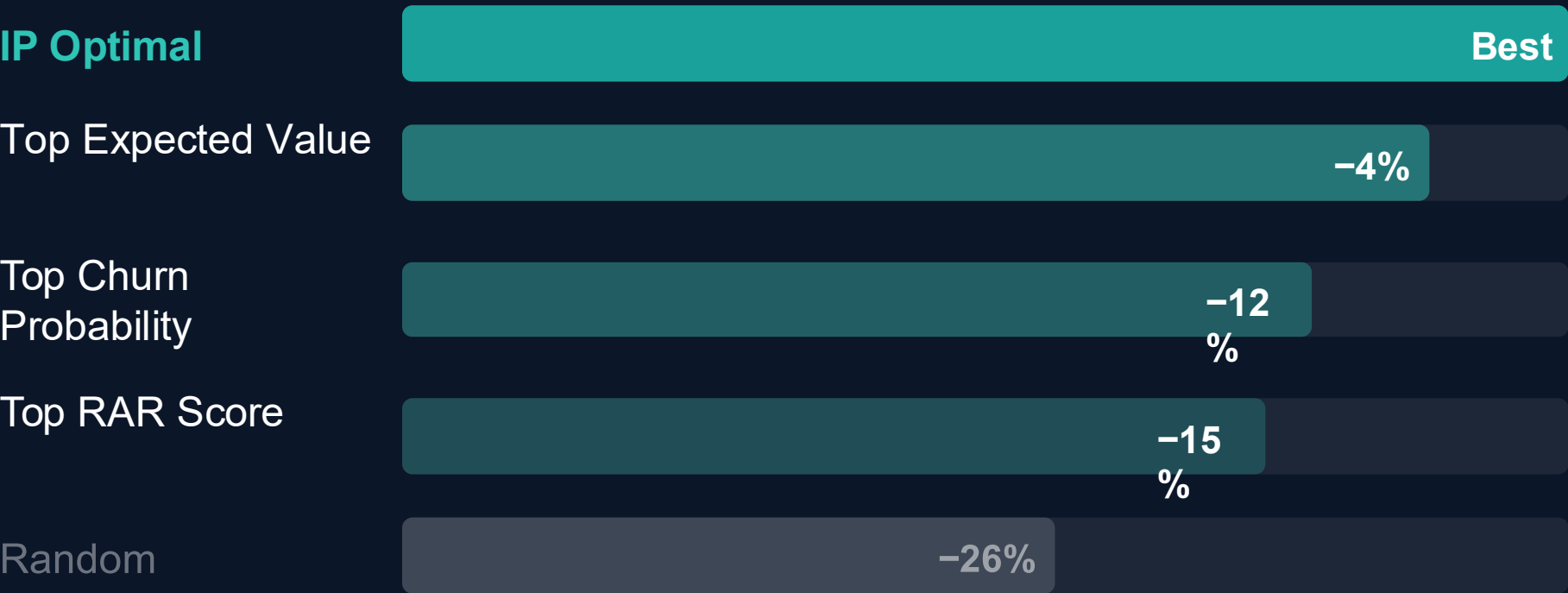
Why Integer Programming?

Each customer can receive only one tier. LP relaxation over-assigns. IP gives the exact optimal allocation — solved with PuLP/CBC in <2 seconds.

OPTIMIZATION RESULTS

IP outperforms every heuristic — with the same budget

Expected Retention Value by Strategy (\$7,500 budget)



IP LIFT VS. RANDOM

+26%

same \$7,500 budget — more value from smarter allocation

TIER ALLOCATION (IP SOLUTION)

Email / SMS	~800 customers
Phone Outreach	~280 customers
Premium Package	~60 customers
Not contacted	Remaining

Cluster Insight

Cluster 0 (highest churn, high CLTV) receives Premium Package at near-100% contact rate.

Recommendation is robust across risk scenarios

How We Model Uncertainty

Each customer's churn probability has variance — estimated from their nearest K-Means cluster (σ_i). We penalize uncertain predictions using a risk aversion parameter **k** (Markowitz-style).

$$v_{ij}^{adj} = (\hat{p}_i - k \cdot \sigma_i) \cdot \eta_j \cdot CLTV_i - c_j$$

- **k = 0 (risk-neutral)**: maximize expected value, no uncertainty penalty
- **k = 1 (moderate)**: slight shift toward lower-variance clusters — recommended operating point
- **k = 2 (conservative)**: favors highly confident predictions; total value drops ~8%
- **Sensitivity to budget**: value scales linearly up to ~\$12K, then diminishing returns as high-CLTV pool is exhausted

Efficient Frontier — Risk Aversion vs. Expected Value



Three-phase rollout over 90 days

DAYS 1–30

Deploy & Score

- Operationalize Random Forest churn scoring pipeline
- Run IP solver monthly with updated churn probabilities
- Pilot Email/SMS tier on Cluster 0 customers ($p_{\text{churn}} > 0.80$)

DAYS 31–60

Expand & Validate

- Roll out Phone Outreach to medium-CLTV / high-churn segments
- Measure actual retention rates vs. η assumptions
- Re-calibrate σ estimates from new outcome data

DAYS 61–90

Optimize & Scale

- Launch Premium Package for top-CLTV churners
- Set $k = 1$ as default risk parameter in production
- Automate monthly re-solve; report ROI dashboard to leadership

DECISION RULE

If $p_{\text{churn}} > 0.80 \rightarrow$ contact immediately. If $v_{ij} > 0 \rightarrow$ assign best tier within budget. Otherwise $\rightarrow$ no action.

RISKS + CLOSING TAKEAWAY

Know the limits — act anyway

η is assumed, not measured

Retention rates (10% / 25% / 45%) are estimates. A/B testing in the pilot phase is critical to validate.

Model drift over time

Churn drivers change with market conditions. Re-train quarterly; monitor feature importance shifts.

CLTV accuracy

CLTV values from the dataset may not reflect current pricing. Overvalued CLTV overspends on the wrong customers.

Descriptive, not causal

The model predicts churn, not intervention response. Customers with high predicted churn may already be in process of leaving.

Don't spend retention budget uniformly.
Spend it where the math says it matters.

+26%

more retention value vs. random targeting

Same \$7,500 budget. Better decisions. Sustained by a risk-robust optimization framework.